MILITARY SPECIFICATION SHEET

CONNECTORS, ELECTRICAL, (CIRCULAR, ENVIRONMENT RESISTING), RECEPTACLE, (SINGLE HOLE MOUNT, THREADED COUPLING, SOLDER PIN CONTACT), (SERIES III, CLASSES H, N, AND Y)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the connector described herein shall consist of this specification and the latest issue of MIL-C-83723.

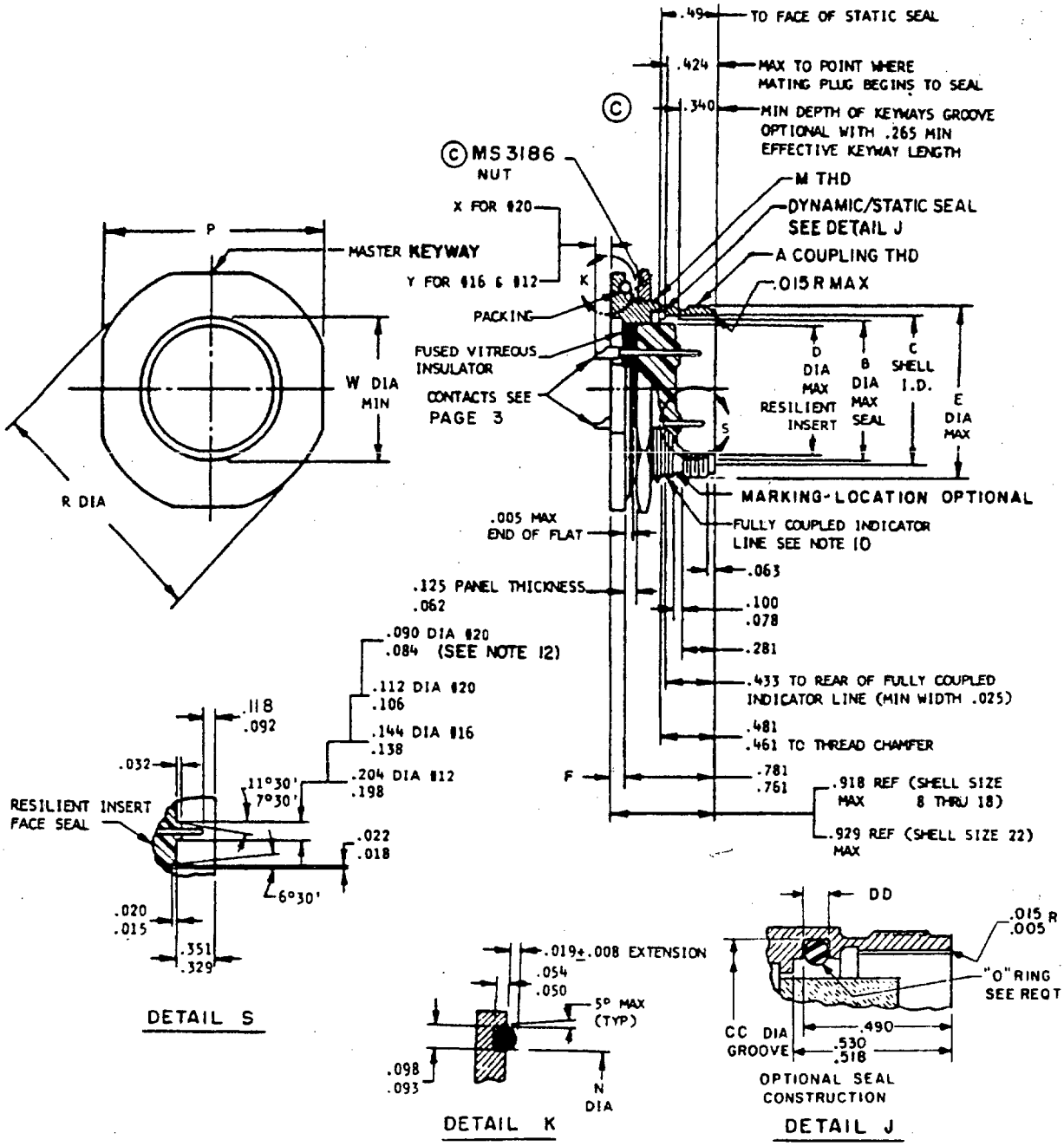


FIGURE 1. Dimensions and configurations.

© denotes changes

AMSC N/A

FSC 5935

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Connector dimensions.

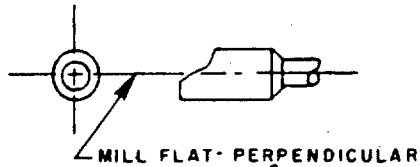
Shell size	A thread			B	C	D	E	F	M thread	N
	Thread	Major dia.	Minor dia.							
8	.562-24 UNEF-2A	.5613 .5541	.5102 .5082	.398	.418 .413	.290	.562	.137 .097	.625-20 UN-2A	.670
10	.688-24 UNEF-2A	.676 .671	.6372 .6280	.519	.535 .530	.388	.696	.137 .097	.750-20 UNEF-2A	.796
12	.875-20 UNEF-2A	.862 .857	.8124 .8043	.689	.705 .700	.558	.875	.137 .097	.938-20 UNEF-2A	.984
14	.938-20 UNEF-2A	.924 .919	.8748 .8666	.758	.774 .769	.627	.936	.137 .097	1.000-20 UNEF-2A	1.046
16	1.062-18 UNEF-2A	1.048 1.043	.9929 .9842	.885	.901 .896	.772	1.062	.137 .097	1.125-18 UNEF-2A	1.171
18	1.188-18 UNEF-2A	1.173 1.168	1.1178 1.1089	.991	1.007 1.002	.860	1.187	.137 .097	1.250-18 UNEF-2A	1.296
22	1.438-18 UNEF-2A	1.423 1.418	1.3678 1.3588	1.241	1.257 1.252	1.110	1.437	.148 .128	1.500-18 UNEF-2A	1.609

(C)

Shell size	P	R	W	X	Y	FF	EE	CC +.005 -.000	DD	JJ
8	.980 .949	1.078 1.047	.400	.180 .120	.210 .150	.635	.605	.490	.076 .071	.596 .590
10	1.104 1.073	1.192 1.162	.411	.180 .120	.210 .150	.760	.730	.636	.089 .084	.721 .715
12	1.291 1.260	1.380 1.349	.581	.180 .120	.210 .150	.947	.917	.806	"	.908 .902
14	1.391 1.360	1.516 1.485	.650	.180 .120	.210 .150	1.010	.980	.875	"	.971 .965
16	1.516 1.485	1.641 1.610	.777	.180 .120	.210 .150	1.135	1.105	1.002	"	1.096 1.090
18	1.641 1.610	1.766 1.735	.864	.180 .120	.210 .150	1.260	1.230	1.108	"	1.220 1.214
22	1.954 1.923	2.078 2.047	1.133	.169 .109	.199 .139	1.510	1.480	1.358	.089 .084	1.470 1.464

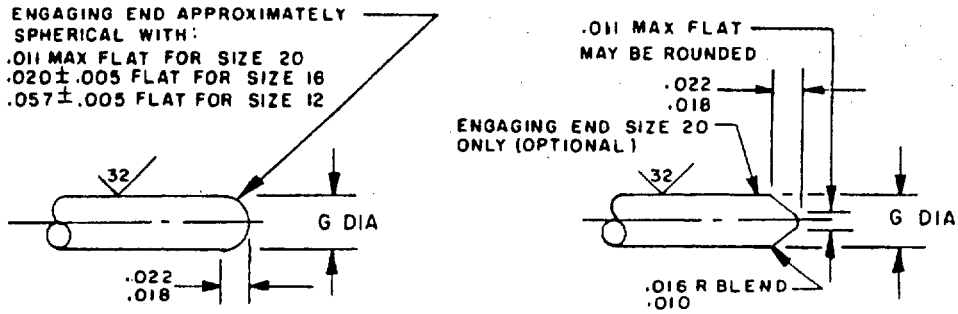
FIGURE 1. Dimensions and configurations - Continued.

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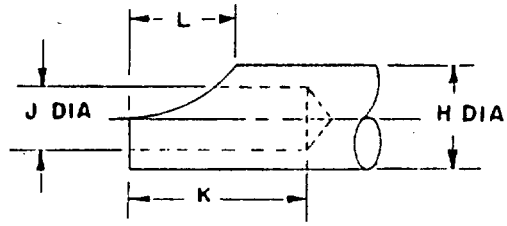


SOLDERPOT ORIENTATION

ENGAGING END OF CONTACTS



TERMINATION END OF CONTACTS



SOLDERPOT

Dimensions of contact engaging end.

Contact size	G	H	J	K	L
20	.041	.088	.048	.188	.110
	.039	.061	.042	.109	.068
16	.0635	.116	.075	.251	.141
	.0615	.096	.069	.172	.109
12	.095	.150	.122	.251	.141
	.093	.130	.112	.172	.109

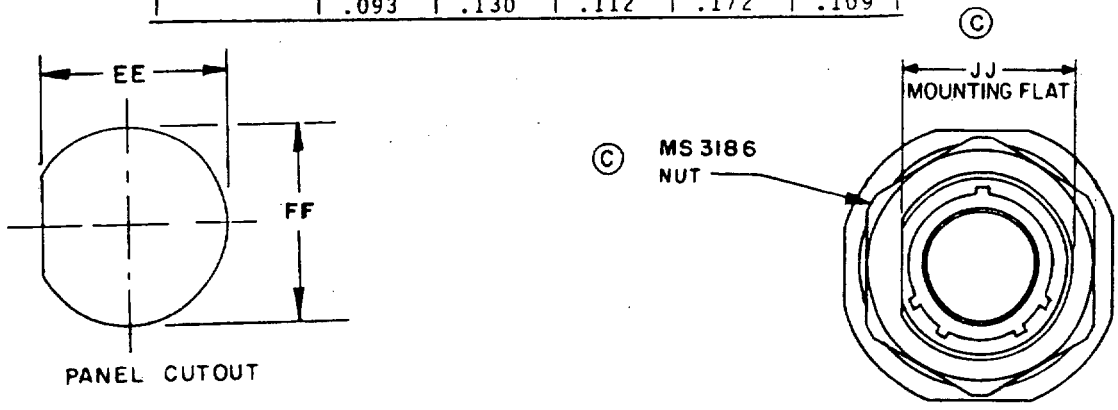


FIGURE 1. Dimensions and configurations - Continued.

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Inches	mm	Inches	mm	Inches	mm	Inches	mm	Inches	mm
.000050	0.00127	.110	2.79	.519	13.18	.8666	22.012	1.162	29.51
.003	0.08	.112	2.84	.530	13.46	.8748	22.220	1.168	29.67
.004	0.10	.116	2.95	.535	13.59	.875	22.23	1.171	29.74
.005	0.13	.118	3.00	.5541	14.074	.885	22.48	1.173	29.79
.008	0.20	.120	3.05	.558	14.17	.896	22.76	1.187	30.15
.010	0.25	.122	3.10	.5613	14.257	.901	22.89	1.192	30.28
.011	0.28	.125	3.18	.562	14.27	.902	22.91	1.205	30.61
.015	0.38	.128	3.25	.581	14.76	.908	23.06	1.214	30.84
.016	0.41	.130	3.30	.590	14.99	.917	23.29	1.220	30.99
.018	0.46	.135	3.43	.596	15.14	.918	23.31	1.230	31.24
.019	0.48	.137	3.48	.605	15.37	.919	23.34	1.241	31.52
.020	0.51	.138	3.51	.627	15.93	.920	23.37	1.252	31.80
.022	0.56	.139	3.53	.6280	15.951	.924	23.47	1.257	31.93
.025	0.64	.141	3.58	.635	16.13	.929	23.60	1.260	32.00
.032	0.81	.144	3.66	.6352	16.134	.936	23.77	1.291	32.79
.039	0.99	.148	3.76	.636	16.15	.947	24.05	1.295	32.89
.041	1.04	.150	3.81	.650	16.51	.949	24.10	1.296	32.92
.042	1.07	.169	4.29	.670	17.02	.954	24.23	1.329	33.76
.048	1.22	.172	4.37	.671	17.04	.965	24.51	1.349	34.26
.050	1.27	.180	4.57	.676	17.17	.971	24.66	1.358	34.49
.054	1.37	.188	4.78	.689	17.50	.980	24.89	1.3588	34.513
.057	1.45	.198	5.03	.696	17.68	.984	24.99	1.360	34.54
.061	1.55	.199	5.05	.700	17.78	.9842	24.999	1.3678	34.742
.0615	1.562	.204	5.18	.705	17.91	.991	25.17	1.380	35.05
.062	1.57	.210	5.33	.708	17.98	.9929	25.220	1.391	35.33
.063	1.60	.251	6.38	.715	18.16	1.0002	25.45	1.418	36.02
.0635	1.613	.265	6.73	.721	18.31	1.0007	25.58	1.421	36.09
.068	1.73	.281	7.14	.730	18.54	1.010	25.65	1.423	36.14
.069	1.75	.290	7.37	.758	19.25	1.043	26.49	1.437	36.50
.071	1.80	.329	8.36	.760	19.30	1.046	26.57	1.455	36.96
.075	1.91	.349	8.64	.761	19.33	1.047	26.59	1.464	37.19
.076	1.93	.351	8.92	.769	19.53	1.048	26.62	1.470	37.34
.078	1.98	.388	9.86	.772	19.61	1.062	26.97	1.480	37.59
.084	2.13	.398	10.11	.774	19.66	1.073	27.25	1.485	37.72
.088	2.24	.400	10.16	.777	19.74	1.078	27.38	1.510	38.35
.090	2.29	.411	10.44	.781	19.84	1.090	27.69	1.516	38.51
.091	2.31	.418	10.62	.795	20.19	1.096	27.84	1.609	40.87
.092	2.34	.424	10.77	.796	20.22	1.104	28.04	1.610	40.89
.093	2.36	.433	11.00	.8043	20.429	1.105	28.07	1.641	41.68
.095	2.41	.461	11.71	.806	20.47	1.108	28.14	1.671	42.44
.096	2.44	.481	12.22	.8124	20.635	1.1089	28.166	1.705	43.31
.098	2.49	.490	12.45	.857	21.77	1.1178	28.392	1.766	44.86
.100	2.54	.5082	12.908	.860	21.84	1.133	28.78	1.923	48.84
.106	2.69	.5102	12.959	.862	21.89	1.135	28.83	1.954	49.63
.109	2.77	.518	13.16	.864	21.95	1.142	29.01	2.047	51.99
								2.078	52.78

FIGURE 1. Dimensions and configurations - Continued.

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NOTES: (Notes 4 through 9 applicable to contacts only).

1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerances are .005 (0.13 mm) for three place decimals and .01 (0.25 mm) for two place decimals. Angles ±1°.
4. Contact finish: Gold plate .000050 inch minimum. In accordance with MIL-G-45204. A suitable underplating shall be used. Silver shall not be used. Finish is applied after sealing.
5. Dimensions apply after plating.
6. 63 all over per ANSI B46. 1-62 unless otherwise specified.
7. Remove all burrs and sharp edges .003 to .005 R.
8. Diameters to be concentric within .004 TIR.
9. Diameters in the same cross-sectional plane to be concentric with each other within .003 TIR.
10. Coupling indicator line shall be color coded blue.
11. Shell size 8 will not mate with MIL-C-26500 connectors.
12. See table I for contact cavities requiring this dimension.

FIGURE 1. Dimensions and configurations - Continued.

TABLE I. Insert arrangements requiring reduced diameters for raised seal barriers on contact cavities as indicated.

Shell size	Insert arrangements	Contact cavities
8	-3	1, 2, and 3

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REQUIREMENTS:

Dimensions and configuration: See figure 1.

This connector mates with MIL-C-83723/86, MIL-C-83723/91, MIL-C-83723/95, and MIL-C-83723/97.

Insert arrangements: See MIL-STD-1554 and table I.

"O" ring: See MIL-C-83723.

Part number example:

M83723/89	H	22	55	N

General specification number

Specification sheet number

Class (see MIL-C-83723)

Shell size (see figure 1)

Insert arrangement (see MIL-STD-1554 and table I)

Polarizing position (see MIL-STD-1554)

Custodians:
Army - CR
Navy - AS
Air Force - 85

Review activities:
Army - AR, AV, MI
Navy - EC
Air Force - 11, 15, 99
DLA - ES

Preparing activity:
Air Force - 85

Agent:
DLA - ES

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